

OWEN MOOGK

Mechatronics Engineering, University of Waterloo

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SKILLS

Languages: Java, C++, Go, Python, C#, SQL (PostgreSQL), R, TypeScript/JavaScript.

Systems and Frameworks: Linux, Docker, GitHub CI/CD, Terraform, FastAPI, Django, React, Next.js.

Specializations: Low latency trading, system architecture, performance profiling, manufacturing execution.

EXPERIENCE

Software Developer (Low-Latency Trading Systems) – Virtu Financial

June 2026 – Present

- Developed and optimized low latency systems in Java for market data aggregation and dissemination to traders.
- Optimized performance of a market data processor by 250% by improving Solarflare Onload TCP configurations, partitioning data sources, and optimizing or removing CPU-bound code from the hot path.
- Reduced latency of an informational market data pipeline from ~1100ms to ~30ms by improving aggregation implementations.
- Improved performance of internal applications by 30% by designing an alternative distributed pub/sub architecture.
- Designed high-availability event bus clients for downtime prevention and halved the recovery time from unexpected outages.
- Profiled performance metrics including network buffering, CPU utilization, and memory consumption to inform optimization and feature development decisions.

Software Developer (Systems & Operations) – Base Power Company

January 2026 – April 2026

- Built a custom cloud-based manufacturing execution system to coordinate production line processes using Golang and PostgreSQL, enabling detailed metrics regarding quality, costs, and operational capability.
- Engineered a bi-directional ERP integration using Temporal workflows to synchronize inventory tracking with message fault-tolerance and idempotence to prevent state machine drift.
- Designed a query builder library with composable filter abstractions, allowing reusable code generation.
- Built end-to-end testing infrastructure with Docker and GitHub Actions CI for fast and reliable feedback on code changes.
- Optimized CI pipelines, reducing runtime by 50% to improve deployment queues and engineering cycle time.
- Managed infrastructure as code across Terraform and AWS to build systems for reliability and uptime.

Software Developer (Internal Platforms) – Rocket Factory Augsburg

September 2024 – December 2024

- Developed operational tools for a 300-person team building advanced rocket technology, using React, Python, and FastAPI.
- Built application features for a manufacturing execution system, streamlining production workflow and tracking key metrics.
- Designed and built a time tracking application used company-wide, reducing administrative overhead by an estimated 70%.
- Utilized CI pipelines to improve testing and deployment of applications, ensuring quality and usability of tools.
- Optimized codebase structure, refactoring and enhancing existing code to enhance performance, reusability, and scalability.
- Improved PostgreSQL ORM indexing to improve hierarchical data traversal and retrieval query times.

Software Developer (Web Platform) – BusPlanner Inc.

May 2023 – September 2023

- Built web applications for a school bus routing system in .NET, deploying features to hundreds of clients across North America.
- Designed platform features and built implementations in C#, JavaScript, and SQL (MSSQL).
- Improved and rebuilt legacy software, optimizing existing code for performance, reusability, and scalability.
- Wrote and performed end-to-end testing, diagnosing and fixing performance and behavior bugs.

EDUCATION

Mechatronics Engineering – University of Waterloo

September 2022 – April 2027

- Candidate for Bachelor of Applied Science studying Mechatronics Engineering, with a grade average of 91%.
- President's Scholarship of Distinction, Douglas Wright Award, International Experience Award, Dean's Honors.
- Relevant courses: Computer Architecture, Real Time Operating Systems, Data Structures and Algorithms, Digital Logic, Microprocessors, Embedded Firmware, Circuits, Advanced Calculus, Linear Algebra, Numerical Methods, Economics.